

Improving omnichannel retail experiences, with Google Kubernetes Engine and Cloud SQL

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT OMNEO

omneo

Omneo uses Google Kubernetes Engine to deliver personalized customer profiles on a scalable microservices architecture and to gain valuable insights into customer sentiment and behavior.

About Omneo

Omneo is a customer experience suite used by retailers across Australia, New Zealand, the United Kingdom, Europe, and the US. The customer data platform is the brainchild of customer experience consultancy Arkade based in Melbourne, Australia.

Industries: Retail & Consumer Goods

Location: Australia

Google Cloud Results

- Reduces 70% of developer time spent on provisioning servers by implementing infrastructure as a code
- Saves 40% of hosting costs by optimizing server resources
- Speeds up delivery of new features and enhancements with automated scaling and deployment

Faster service market running development

servian

About Servian

Servian is a Google Cloud Partner specializing in data analytics, application development, marketing analytics, and machine learning.

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Cloud AI & Machine Learning
(<https://cloud.google.com/products/ai/>)

Cloud Memorystore
(<https://cloud.google.com/memorystore/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

The customer is at the center of any retail business.

Omneo (<https://www.omneo.io/>) helps retailers to understand their customers and create personalized experiences with data and analytics. Hosted on Google Cloud (<https://cloud.google.com/>), the Omneo Customer Experience (CX) suite connects and analyzes customer data from multiple channels

such as POS system, email subscription, and customer service software, to create a holistic view of the customer's journey, offline and online.

Retailers add a personal touch by looking up customer preferences such as payment method or favorite store through Omneo's customer relationship management portal, Clienteling, hosted on [App Engine](#) (/appengine). Omneo helps retailers extend their in-store reach with branded mobile apps built on [Google Kubernetes Engine](#) (/kubernetes-engine), such as Profile Portal which allows customers to maintain their profile, track purchases, and redeem rewards at any time.

After an initial launch in 2010 with only three clients, followed by two complete rebuilds in 2015, Omneo's consumer reach has grown tenfold. It was becoming a significant challenge to deliver up-to-date profile and incentive data to customers across brands, locations, and channels, so the company turned to Google Cloud for a solution.

"Developers used to spend three weeks a month on product updates," says Danny Phillips, Co-Founder at Omneo. "By deploying microservices on Google Kubernetes Engine, we're rolling out dozens and even hundreds of features and updates each month, such as digital loyalty cards and passwordless authentication."

"Google Cloud helps us fast-track migration of all clients onto a centralized infrastructure without compromising on data availability, security, integration practices, or application modularity and

support,” says Drew Morris, Product Architect at Omneo.

Omneo enlisted the expertise of Google Cloud Partner [Servian](https://www.servian.com/) to re-architect its core infrastructure and established Google Cloud DevOps best practices to reduce the operational burden on its developers.

“With the help of Servian, we migrated to Google Cloud in eight weeks with minimum developer input and maximum performance output,” says Danny.

“Google Cloud gives us more time to focus on building customer-facing product features rather than managing infrastructure tasks.”

[Omneo] Offering secure services t...



“The native support for Kubernetes in Google Cloud gives us the confidence to containerize our services and run them all within the same cluster. Instead of running one application per server, we can optimize servers with multiple applications and reduce hosting costs by 40%.”

—Danny Phillips, Co-
Founder, Omneo

**Scaling cost-effectively to meet
customer demands**

In the past, Omneo developed each project on client-managed infrastructure. Omneo consolidated all applications on Google Cloud to improve developer workflow. Using Google Cloud as the hosting environment, Omneo internalizes hosting costs and removes the infrastructure burden from the customers so they can focus on their core business.

“The native support for Kubernetes in Google Cloud gives us the confidence to containerize our services and run them all within the same cluster,” says Drew. “Instead of running one application per server, we can optimize servers with multiple applications and reduce hosting costs by 40%.”

Omneo uses microservices to power all of the third-party integrations such as ecommerce apps and loyalty partner programs as well as Profile Portal as its separate service.

Faced with an ever-growing product suite, Omneo needed a solution to deliver bespoke or ad hoc applications that didn't fit into its existing range of products. App Engine gives Omneo the ability to manage the development of core products and client-specific applications in one centralized environment.

App Engine supports multi-tenancy, so Omneo can bill customers for resources specific to domains in their accounts. This way, Omneo's base operating costs aren't skewed by charges incurred by partners or customers.

Engaging customers with real-time, personalized insights

On any given day, Omneo experiences more than 10,000 customer events per brand per minute. These interactions range from new transactions and product returns to rewards creation and redemption. Omneo's legacy infrastructure struggled to keep up with customer queries during high-traffic sales events.

"Moving to fully managed database service Cloud SQL enables us to have a real-time view of customer interactions across channels," says Drew. "We no longer face lengthy delays between a customer joining or transacting and the availability of that data in Omneo and all other systems downstream."

Omneo uses Memorystore (/memorystore) to queue tasks that need to happen at different times in the event journey. When a redemption happens, for example, it processes this event and stores the result in Cloud SQL, then queues it up in Memorystore to notify other application components that a redemption has happened.

"While some brand and marketing teams are happy to have end-of-day data freshness for reporting and insights, consumers, and those that serve them, need everything to be personal, and in real time. Omneo is built for them first," says Danny.

Omneo also helps retailers improve decision-making by reporting key customer metrics from its application. The monthly snapshot captures data

such as new customer momentum, the value generated by customers in their first three months after joining the program, and customer churn rate, the percentage at which customers have canceled their subscription.

“Moving to fully
managed
database service
Cloud SQL
enables us to
have a real-time
view of
customer
interactions
across
channels.”

—Drew Morris, Product
Architect, Omneo

Improving developer efficiency by automating infrastructure tasks

Transitioning to Google Cloud reduces the maintenance burden for Omneo’s DevOps teams. It

now takes minutes instead of days to provision servers for new client environments. Omneo uses third-party solutions that integrate easily with Google Cloud, such as Terraform to manage infrastructure as a code and CircleCI to automate testing and deployment.

Omneo requires minimal DevOps resources to manage its large and growing infrastructure by deploying Terraform self-service clusters on Google Kubernetes Engine. This architecture allows Omneo to scale web servers independently and cost-efficiently instead of over-provisioning infrastructure.

Omneo executes Terraform on CircleCI to test each code before applying changes in its production environment. This approach enables Omneo to maintain its desired state of infrastructure and minimize downtime caused by bad code. Running a series of unit and integration tests helps to identify issues across a vast breadth of functionality.

“By deploying
microservices
on Google
Kubernetes
Engine, we’re
rolling out
dozens and even

hundreds of
features and
updates each
month, such as
reward
extensions and
recent profile
views on Profile
Portal.”

—*Danny Phillips, Co-
Founder, Omneo*

Minimizing cyber risks with data encryption and user permission

Data in Google Cloud is encrypted at rest and in transit to reduce data loss. Each customer database is kept separately in its own Google Cloud project so there’s no risk of data comingling between Omneo’s customer accounts.

Omneo uses Identity and Access Mangement (/iam) to control who has access to infrastructure resources on Google Cloud. Omneo sets minimum permissions possible to give project members the required access to finish the tasks at hand.

“Omneo respects consumer privacy by design. We treat our client’s consumer data with the utmost respect,” says Danny. “It’s important to us that

Google Cloud complies with rigorous privacy and security standards to handle personally identifiable and sensitive data.”

Moving forward, Omneo wants to harness the power of BigQuery for analytics and Cloud AI for AI and machine learning to help retailers improve customer engagement. Using predictive analytics, Omneo can anticipate customer needs and offer hyper-personal recommendations that appeal to the customer exactly when they need it.